# A Proof that $a(n)/n^2 \to \frac{1}{2}$ for OEIS Sequence A112509

Ben Clare

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**Abstract**

For $n \geq 1$, let $a(n)$ denote the maximum number of distinct non-negative integers whose binary representations appear as contiguous substrings of some $n$-bit binary string (OEIS sequence A112509). This note proves that

$$\lim_{n\to\infty} \frac{a(n)}{n^2} = \frac{1}{2}.$$

The upper bound $a(n) \leq n(n+1)/2$ is elementary. For the matching lower bound an explicit family of $n$-bit strings is exhibited achieving at least $\frac{1}{2}n^2 - O(n^{5/3})$ distinct integer values. The argument is self-contained and elementary.

## 1 Introduction

Given a binary string $s$ of length $n$, every contiguous substring can be read as a non-negative integer in binary (with leading zeros ignored in the standard way: $\texttt{01} = \texttt{1} = 1$, $\texttt{00} = \texttt{0} = 0$).

**Definition 1.** For a binary string $s \in \{0,1\}^n$, let $f(s)$ denote the number of distinct non-negative integers whose binary representations appear among the contiguous substrings of $s$. Define

$$a(n) = \max_{s\in\{0,1\}^n} f(s).$$

The sequence $a(n)$ is listed as OEIS A112509 [1]. It begins

$$1, 3, 5, 7, 10, 13, 17, 22, 27, 33, 40, 47, 55, 64, 73, 83, 94, 106, 118, 131, \ldots$$

Prior computational work [2] established exact values for $n \leq 150$ and certified lower bounds up to $n = 2 \times 10^9$ via suffix-array-based search. Selected values:

| $n$ | $a(n)$ or lower bound | $a(n)/n^2 \geq$ |
|---:|---:|---|
| 50 | 899 | 0.3596 |
| 100 | 3,875 | 0.3875 |
| 150 | 9,070 | 0.4031 |
| $10^6$ | 498,396,662,200 | 0.4984 |
| $10^9$ | 499,943,911,277,037,650 | 0.49994 |
| $2 \times 10^9$ | 1,999,841,302,297,432,600 | 0.49996 |

The purpose of this note is to prove the limit rigorously.

**Theorem 2.** $\lim_{n\to\infty} \dfrac{a(n)}{n^2} = \dfrac{1}{2}$.

The proof combines an elementary upper bound (Section 2) with an explicit lower-bound construction (Sections 3 –5).

# 2 Upper bound

**Lemma 3** (Upper bound)**.** *For all $n \geq 1$, $a(n) \leq \dfrac{n(n+1)}{2}$.*

*Proof.* Two substrings of $s$ represent the same integer if and only if they are identical after stripping leading zeros. Consequently every distinct positive integer value corresponds to a unique substring beginning with `1`.

*Case 1.* Suppose $s$ contains at least one `0` bit. Then the value 0 is also represented, but only by substrings beginning with `0`. At least one of the $\frac{n(n+1)}{2}$ substrings begins with `0`, so the count of leading-`1` substrings is at most $\frac{n(n+1)}{2} - 1$. Adding the single value 0 gives $f(s) \leq \frac{n(n+1)}{2}$.

*Case 2.* If $s$ is the all-ones string then $f(s)$ equals the number of distinct positive values, which is at most $\frac{n(n+1)}{2}$.

Hence $a(n) \leq \frac{n(n+1)}{2}$ and $\limsup_{n\to\infty} a(n)/n^2 \leq \frac{1}{2}$. □

*Remark* 4. The bound $a(n) \leq n(n+1)/2$ is tight at $n = 2$: the string `11` has substrings `1`, `1`, `11` with values $1, 1, 3$, giving $f(\texttt{11}) = 2 = 2 \cdot 3/2 - 1$. Wait — `10` gives substrings with values $\{0, 1, 2\}$, so $a(2) = 3 = 2 \cdot 3/2$. The bound is achieved.

# 3 The construction

Fix $n \geq 8$. Set

$$m = \lfloor n^{1/3} \rfloor, \qquad C = \frac{m(m-1)}{2}, \qquad L = \left\lfloor \frac{n-C}{m} \right\rfloor, \qquad r = n - C - mL.$$

Since $L = \lfloor (n-C)/m \rfloor$ we have $0 \leq r < m$ and $n = mL + C + r$. Note that $m \geq 2$ for $n \geq 8$, and $L \geq (n-C)/m - 1 \geq n/(2m) \to \infty$, so $L \geq 1$ for all sufficiently large $n$ (in fact for all $n \geq 8$).

Define the $n$-bit binary string

$$s_n \;=\; \underbrace{1^L}\, 0^1\, \underbrace{1^L}\, 0^2\, \underbrace{1^L}\, 0^3 \cdots \underbrace{1^L}\, 0^{m-1}\, \underbrace{1^{L+r}} \tag{1}$$

consisting of $m$ one-blocks. The first $m-1$ one-blocks each have length $L$; the final one-block has length $L + r$. The zero-separator lengths are $1, 2, 3, \ldots, m-1$ in order.

**Lemma 5** (Length check)**.** *The string $s_n$ has length $n$.*

*Proof.* The total length is $(m-1)L + (L+r) + (1+2+\cdots+(m-1)) = mL + r + C = n$. □

# 4 A large injective family of substrings

For integers $i, j, \alpha, \beta$ satisfying

$$1 \le i < j \le m, \qquad 1 \le \alpha \le L, \qquad 1 \le \beta \le L,$$

define the substring $u_{i,j,\alpha,\beta}$ of $s_n$ obtained by starting inside the $i$-th one-block at the position where exactly $\alpha$ ones remain in that block, and ending inside the $j$-th one-block immediately after its first $\beta$ ones.

Because each intermediate one-block (blocks $i+1$ through $j-1$) has length exactly $L$ and $\beta \le L \le L + r$ (so the endpoint lies inside the $j$-th block regardless of whether $j = m$), the substring is well-defined. Explicitly, its binary form is

$$u_{i,j,\alpha,\beta} \;=\; \underbrace{1^{\alpha}}_{\text{tail of block } i}\;\underbrace{0^{i}}_{\text{sep.}}\;\underbrace{1^{L}\,0^{i+1}}_{\text{block } i+1}\;\underbrace{1^{L}\,0^{i+2}}_{\text{block } i+2}\cdots\underbrace{1^{L}\,0^{j-1}}_{\text{block } j-1}\;\underbrace{1^{\beta}}_{\text{head of block } j}. \tag{2}$$

Every such word begins and ends with `1`, so it is already the standard binary expansion of a positive integer.

**Lemma 6** (Injectivity)**.** *The map $(i, j, \alpha, \beta) \mapsto u_{i,j,\alpha,\beta}$ is injective: distinct quadruples yield distinct binary strings.*

*Proof.* Let $(i, j, \alpha, \beta)$ and $(i', j', \alpha', \beta')$ be two quadruples with $u_{i,j,\alpha,\beta} = u_{i',j',\alpha',\beta'}$ as binary strings. Since both strings begin and end with `1`, their run-length encodings are uniquely determined. From (2), the run-length encoding of $u_{i,j,\alpha,\beta}$, listing alternating one-block and zero-block lengths, is

$$(\alpha,\ i,\ L,\ i+1,\ L,\ i+2,\ \ldots,\ L,\ j-1,\ \beta).$$

Equality of run-length encodings gives:

- First entry: $\alpha = \alpha'$.
- Last entry: $\beta = \beta'$.
- First zero-block length (second entry): $i = i'$.
- Last zero-block length: $j - 1 = j' - 1$, so $j = j'$.

Hence $(i, j, \alpha, \beta) = (i', j', \alpha', \beta')$, proving injectivity. □

# 5 Lower bound and asymptotics

**Lemma 7** (Counting)**.** $a(n) \ge \binom{m}{2} L^2$.

*Proof.* There are $\binom{m}{2}$ choices for the pair $(i, j)$ and $L^2$ choices for $(\alpha, \beta)$. By Lemma 6, the $\binom{m}{2}L^2$ corresponding substrings are pairwise distinct binary strings, each beginning with `1`, and hence represent pairwise distinct positive integers. They are all substrings of $s_n$, so $f(s_n) \ge \binom{m}{2}L^2$ and thus $a(n) \ge \binom{m}{2}L^2$. □

Theorem 2 now follows.

*Proof of Theorem 2.* The upper bound $\limsup_{n\to\infty} a(n)/n^2 \le \frac{1}{2}$ is Lemma 3. For the lower bound, use Lemma 7 together with $L \ge (n-C)/m - 1 = (n-C-m)/m$:

$$a(n) \;\ge\; \binom{m}{2} L^2 \;\ge\; \frac{m(m-1)}{2} \cdot \left(\frac{n-C-m}{m}\right)^2 \;=\; \frac{m-1}{2m}\,(n-C-m)^2.$$

Dividing by $n^2$:

$$\frac{a(n)}{n^2} \;\ge\; \frac{1}{2}\left(1-\frac{1}{m}\right)\left(1-\frac{C+m}{n}\right)^2. \tag{3}$$

Now $C+m = m(m-1)/2 + m = m(m+1)/2$. Since $m = \lfloor n^{1/3} \rfloor$, we have $m \to \infty$ and

$$\frac{C+m}{n} = \frac{m(m+1)/2}{n} = O\left(\frac{m^2}{n}\right) = O\big(n^{-1/3}\big) \to 0.$$

Therefore the right-hand side of (3) tends to $\frac{1}{2}\cdot 1 \cdot 1 = \frac{1}{2}$, giving $\liminf_{n\to\infty} a(n)/n^2 \ge \frac{1}{2}$.

Combining with the upper bound completes the proof. □

# 6 Quantitative corollary

**Corollary 8.** *For all $n \ge 8$,*

$$a(n) \;\ge\; \frac{n^2}{2} - O\big(n^{5/3}\big)\,.$$

*Proof.* Starting from $a(n) \ge \frac{m-1}{2m}(n-C-m)^2$, expand $(n-C-m)^2 = n^2 - 2n(C+m) + (C+m)^2$. Since $C+m = m(m+1)/2 = O(m^2) = O(n^{2/3})$ and $m = O(n^{1/3})$:

$$(n-C-m)^2 = n^2 - 2n\cdot O(n^{2/3}) + O(n^{4/3}) = n^2 - O(n^{5/3}),$$
$$\frac{m-1}{2m} = \frac{1}{2} - \frac{1}{2m} = \frac{1}{2} - O(n^{-1/3}).$$

Thus

$$a(n) \;\ge\; \big(\tfrac{1}{2} - O(n^{-1/3})\big)\,\big(n^2 - O(n^{5/3})\big) \;=\; \frac{n^2}{2} - O(n^{5/3}).$$ □

*Remark* 9. The error term $O(n^{5/3})$ is an artefact of the simple construction. Computational experiments using suffix-array-guided local search yield strings achieving ratios within 0.004% of $\frac{1}{2}$ at $n = 2\times 10^9$, far closer to the upper bound, but those refined constructions are not needed for the asymptotic theorem.

# 7 A worked example

**Example 10** ($n = 27$)**.** With $n = 27$ one computes $m = \lfloor 27^{1/3} \rfloor = 3$, $C = 3$, $L = \lfloor 24/3 \rfloor = 8$, $r = 0$. The construction gives

$$s_{27} = \underbrace{11111111}_{8}\;0\;\underbrace{11111111}_{8}\;00\;\underbrace{11111111}_{8},$$

a 27-bit string of the form $\mathtt{1}^8\mathtt{0}^1\mathtt{1}^8\mathtt{0}^2\mathtt{1}^8$.

The three pairs $(i,j)$ with $1 \le i < j \le 3$ are $(1,2)$, $(1,3)$, and $(2,3)$. For each pair there are $L^2 = 64$ choices of $(\alpha,\beta)$, giving $3\cdot 64 = 192$ distinct substrings:

- $(i, j) = (1, 2)$: strings of the form $\mathtt{1}^\alpha\mathtt{01}^\beta$ (spanning the first separator).
- $(i, j) = (1, 3)$: strings of the form $\mathtt{1}^\alpha\mathtt{01}^8\mathtt{001}^\beta$ (spanning both separators).
- $(i, j) = (2, 3)$: strings of the form $\mathtt{1}^\alpha\mathtt{001}^\beta$ (spanning the second separator).

These three families are distinguished by their zero-block structure (one zero, then eight ones then two zeros, then two zeros, respectively), confirming injectivity directly. Hence $a(27) \geq 192$. The exact value is $a(27) = 246$ (easily verified by exhaustive enumeration of all $2^{26}$ strings of length 27), so the bound is not tight but the asymptotic argument is correct.


# Acknowledgements

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# References


[1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, Sequence A112509, `https://oeis.org/A112509`, 2026.

[2] B. Clare, *A112509: computational exploration of OEIS sequence A112509*, `https://github.com/axelsmagichammer/A112509`, 2026.